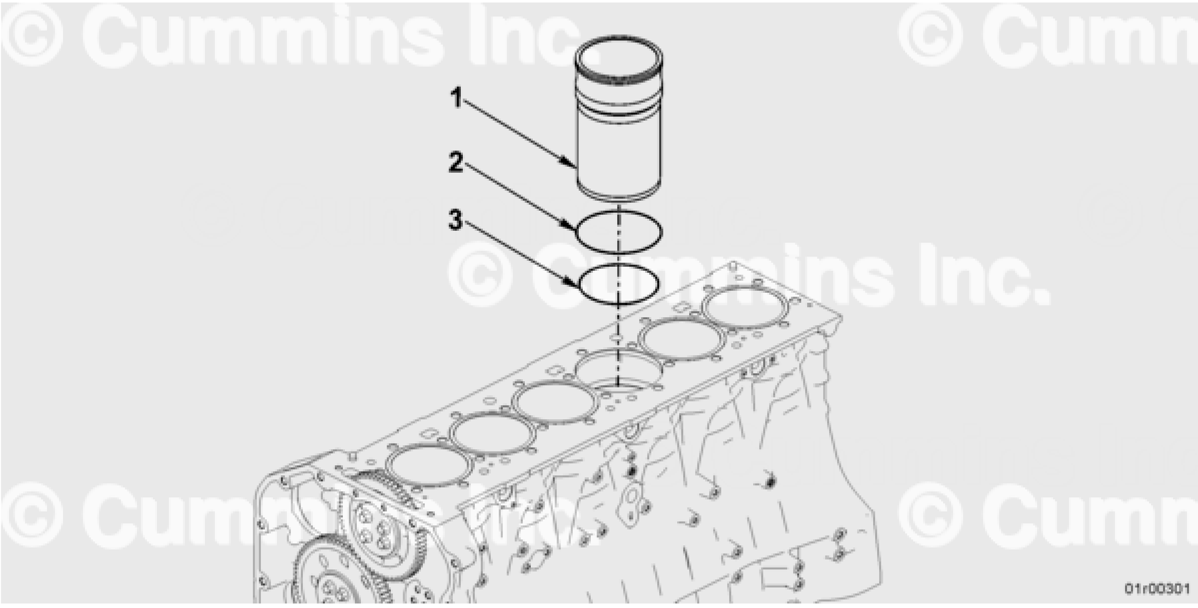
Trainings

Exploded View



Cylinder Liner Exploded View

1. Cylinder liner - quantity 1
2. O-ring - quantity 1
3. O-ring - quantity 1

Select Service Tools

Recommended Cummins® Service Tools

- Depth gauge, Part Number 5299194
- Cylinder Liner Puller Kit, Part Number 5572914
- Scotch-Brite™ 7448, Part Number 3823258, or equivalent

Additional Service Items

- Dykem™, or equivalent
- Soft wire brush
- Dial bore gauge.

Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [122°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

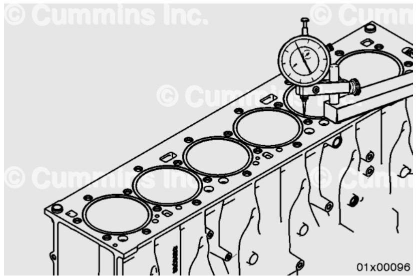
Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

Cummins Inc. does **not** recommend removing the cylinder liners to reduce oil consumption unless the cylinder liners are damaged and **must** be replaced.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/1016/1016-008-018.html)
- Drain the lubricating oil system. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/1016/1016-007-037.html)
- Remove the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/1016/1016-002-004.html)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/1016/1016-007-025.html)
- Remove the block stiffener plate. Refer to Procedure 001-089 in Section 1. (/qs3/pubsys2/xml/en/procedures/1016/1016-001-089.html)
- Remove the piston cooling nozzles. Refer to Procedure 001-046 in Section 1. (/qs3/pubsys2/xml/en/procedures/1016/1016-001-046.html)
- Remove the piston and connecting rod assemblies. Refer to Procedure 001-054 in Section 1. (/qs3/pubsys2/xml/en/procedures/1016/1016-001-054.html)

Initial Check

The cylinder liner protrusion measurement does **not** need to be taken with the cylinder liner in the clamped state.

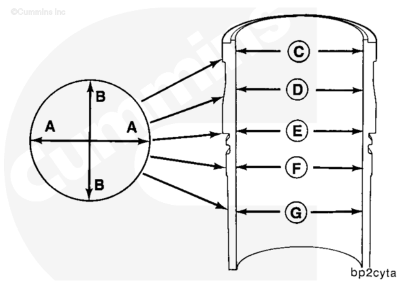


Use depth gauge, Part Number 5299194, to measure the liner protrusion at four points 90 degrees apart.

Zero the depth gauge on the cylinder block before sliding onto the cylinder liner to get the protrusion measurement.

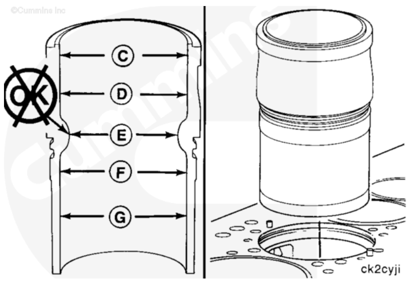
Cylinder Liner Protrusion		
mm		in
0.07	MIN	0.003
0.13	MAX	0.005

Liner Roundness



- While the liner is installed, measure the liner bore for out-of-roundness at points C, D, E, F, and G.
    - For anti-polishing ring liners, take measurement C immediately below the anti-polishing ring counterbore.
- Measure each point in the direction AA and BB.
- The bore **must not** exceed the out-of-round specification.

Liner Bore Out-of-Round		
mm		in
0.03	MAX	0.001

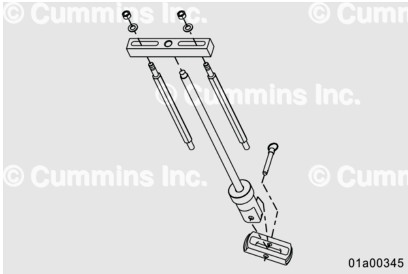


- If the liner bore exceeds the out-of-round specification, remove the liner so that the cylinder block liner bore can be measured.

The block counterbore diameters above and below the cylinder block counterbore area are **not** critical dimensions and do **not** need to be measured.

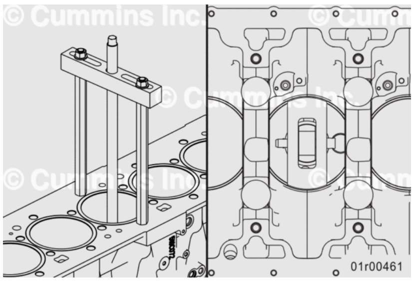
Remove

The cylinder liners can be removed using Cylinder Liner Puller Kit, Part Number 5572914.



⚠ CAUTION ⚠

The cylinder liner puller must be installed and used as described to reduce the possibility of damage to the cylinder block. The puller plate must be parallel to the main bearing saddles and must not extend past the cylinder liner outside circumference.

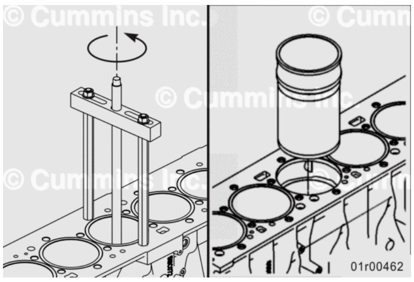


Install the cylinder liner puller onto the top of the cylinder block. Align the bottom of the cylinder liner with the cylinder liner puller.

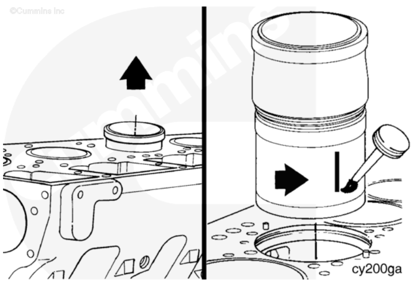
The cylinder liner puller **must** be centered on the top of the cylinder block.

Turn the puller screw stem **counterclockwise** to pull the cylinder liner from the cylinder block.

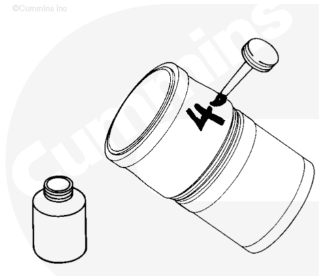
Use both hands to remove the cylinder liner.



When the liner is removed from the cylinder block, use Dykem™, or equivalent, to place a mark on the camshaft side of the cylinder liner to show cylinder liner orientation.



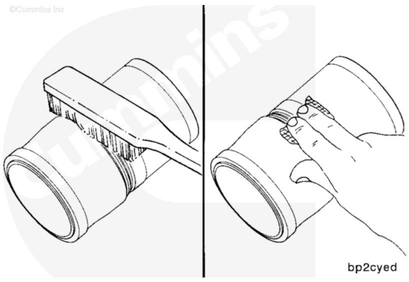
Use Dykem™, or equivalent, to mark the cylinder number on each cylinder liner.



Clean and Inspect for Reuse

⚠ CAUTION ⚠

Do not use an emery cloth or sandpaper to remove carbon from the cylinder liners. Aluminum oxide or silicon particles from emery cloth or sand paper can cause serious engine damage. Do not use any abrasives in the ring travel area. The cylinder liner can be damaged.

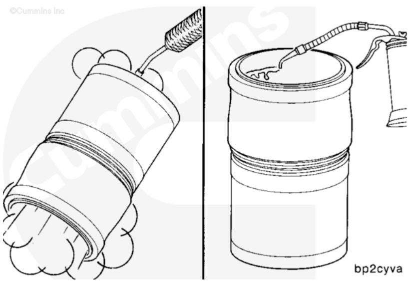


Use a soft wire brush to clean the flange seating area.

Use a fine, fibrous, abrasive pad such as Scotch-Brite™ 7448, Part Number 3823258, or equivalent, to remove the remaining scale and rust.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

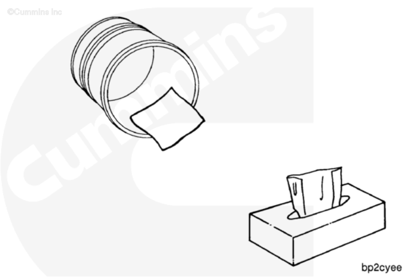
Use solvent or steam to clean the liners. Dry with compressed air.

Use clean engine oil to lubricate the inside circumference of the liners.

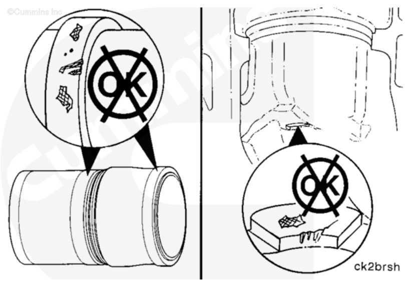
Allow the oil to soak into the liners for 5 to 10 minutes.

Use lint-free paper towels to wipe the excess oil from the inside of the cylinder liners.

Continue to lubricate the inside of the cylinder liners and wipe them clean until the paper towel shows no gray or black residue.

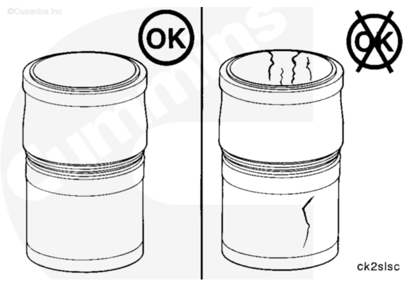


Inspect the liners and cylinder block for dirt or damage.



Inspect the liners for cracks on the inside and outside circumferences.

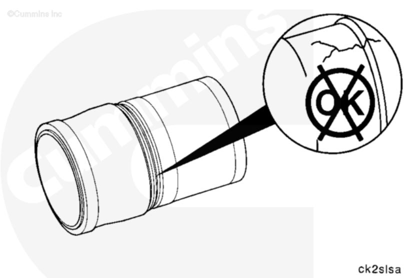
If cracks are found, the cylinder liner **must** be replaced.



Inspect for cracks under the cylinder liner flange.

Cracks can also be detected using either magnetic inspection or the dye penetrant method.

If cracks are found, the liner **must** be replaced.

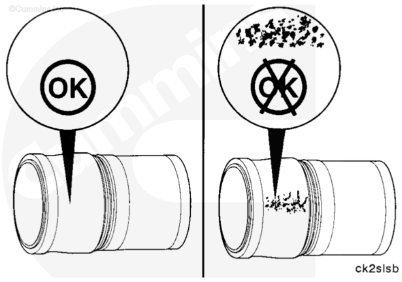


Inspect the outside circumference for excessive corrosion or pitting.

Cylinder liners with pitting generally can **not** be used.

However, if the pitting is light and can be removed with fine emery cloth, the cylinder liner can be used.

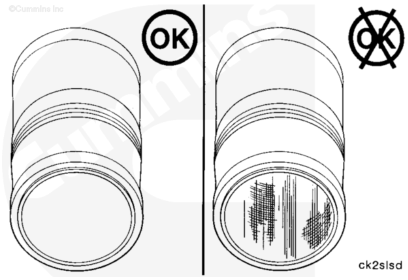
Cylinder Liner Pitting Depth		
mm		in
1.30	MAX	0.051



Inspect the inside surface for vertical scratches deep enough to be felt with a fingernail.

If a fingernail catches in the scratch, the cylinder liner **must** be replaced.

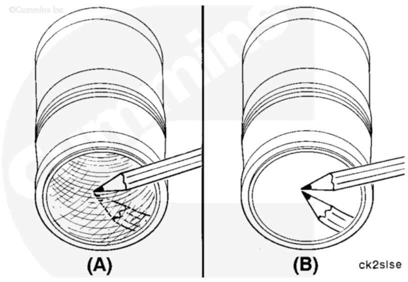
Inspect the inside surface for scuffing or scoring.



Inspect the inside surface for cylinder liner bore polishing.

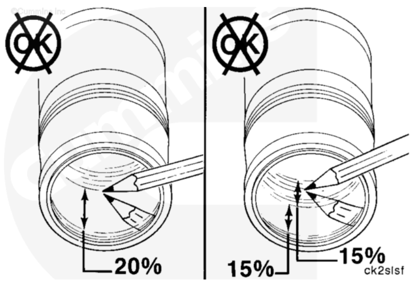
A moderate polish (A) produces a mirror finish in the worn area with traces of the original hone marks or an indication of an etch pattern.

A heavy polish (B) produces a bright mirror finish in the worn area with no traces of hone marks or an etch pattern.



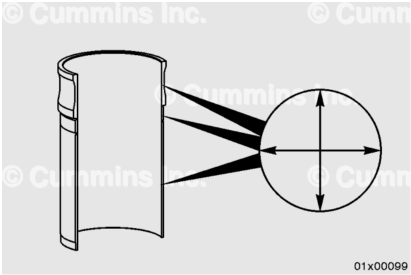
Replace the cylinder liner if:

- A heavy polish is present over 20 percent of the piston ring travel area.
- 30 percent of the piston ring travel has both moderate and heavy polish and one half (15 percent) is heavy polish.



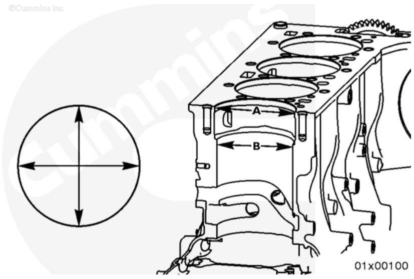
Measure

Use a dial bore gauge to measure the liner inside diameter in four places, 90 degrees apart, at the top and bottom of the piston travel area.



Cylinder Liner Inside Diameter		
mm		in
135.00	MIN	5.315
135.03	MAX	5.316

Measure the cylinder block upper liner bore (A).

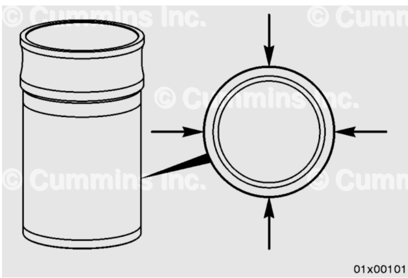


Cylinder Block Upper Liner Bore Inside Diameter (A)		
mm		in
151.275	MIN	5.956
151.325	MAX	5.958

Measure the cylinder block liner seal seat bore (B) 8 mm [0.31 in] to 13.5 mm [0.53 in] below the counterbore.

Cylinder Block Liner Seal Seat Bore Inside Diameter (B)		
mm		in
143.97	MIN	5.668
144.01	MAX	5.670

Measure the liner outside diameter (A).



Cylinder Liner Press Fit Outside Diameter (A)		
mm		in
143.985	MIN	5.669
144.015	MAX	5.670

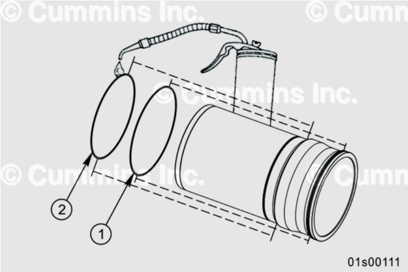
The cylinder block liner counterbore flange diameter is **not** a critical dimension and does **not** need to be measured.

Install

Make sure the cylinder block and all parts are clean before assembly.

Use clean oil to coat the liner o-ring seals.

The color of upper o-ring seal (1) is orange and the color of lower o-ring seal (2) is black.



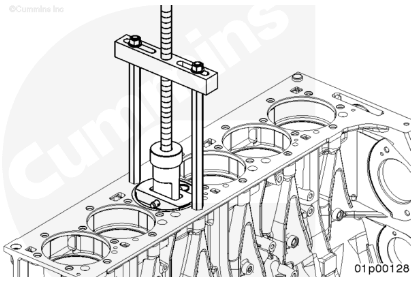
Install the liner into the cylinder block.

When acceptable used liners are installed, rotate the liners 90 degrees from their original position in the engine. Original thrust and anti-thrust surfaces **must** face the front and back of the cylinder block.

Use Cylinder Liner Puller Kit, Part Number 5572914, or equivalent, to drive the liner into the cylinder block bore.

If the liner does **not** seat properly, remove the liner. Inspect the counterbore seat and liner for nicks, burrs, and dirt. Clean if needed.

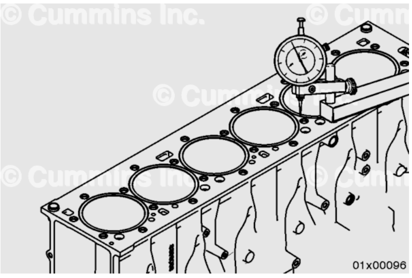
Install the liner again.



The cylinder liner protrusion **must** be checked before installing the pistons and connecting rods.

Use depth gauge, Part Number 5299194, to measure the liner protrusion at four points 90 degrees apart.

Cylinder Liner Protrusion		
mm		in
0.07	MIN	0.003
0.13	MAX	0.005



Finishing Steps

⚠ **WARNING** ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ **WARNING** ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Install the piston and connecting rod assemblies. Refer to Procedure 001-054 in Section 1. (/qs3/pubsys2/xml/en/procedures/1016/1016-001-054.html)
- Install the piston cooling nozzles. Refer to Procedure 001-046 in Section 1. (/qs3/pubsys2/xml/en/procedures/1016/1016-001-046.html)
- Install the block stiffener plate. Refer to Procedure 001-089 in Section 1. (/qs3/pubsys2/xml/en/procedures/1016/1016-001-089.html)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/1016/1016-007-025.html)
- Install the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/1016/1016-002-004.html)
- Fill the lubricating oil system. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/1016/1016-007-037.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/1016/1016-008-018.html)
- Operate the engine to normal operating temperature and check for proper operation and leaks.

All engines **must** be run-in after a rebuild or a repair involving the replacement of one or more piston ring sets, cylinder liners, or pistons. Use one of the following two procedures to complete the run-in procedure.

- Refer to Procedure 014-004 in Section 14. (/qs3/pubsys2/xml/en/procedures/1016/1016-014-004.html)
- Use the following procedure for general run-in test overview. Refer to Procedure 014-999 in Section F. (/qs3/pubsys2/xml/en/procedures/1016/1016-014-999.html)

Last Modified: 03-Apr-2024